

The Finite Element Method

Finite Element PLANE4

Christian Schenk

Master's program in Civil Engineering

Thursday 13th November, 2025

Abstract

In the following, the implementation of the finite element method for the isoparametric quadrilateral element with four nodes (PLANE4) is shown. A 2-D static analysis is carried out (plane stress state). The material is assumed to be linearly elastic and isotropic.

1 Problem description

Given is a tip loaded cantilever plate, see Fig. 2 and example 4.9 in Bathe [1]. The mesh, node and element numbering, respectively, is shown in Fig. 2. The quadratic plate has the length of 4 cm and a thickness of 0.1 cm (Young's modulus of 2×10^5 N/mm² and Poisson's ratio of 0.3).

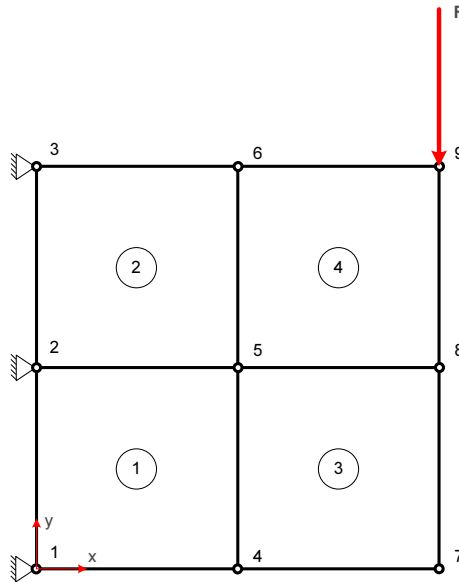


Figure 1: Mesh, node and element numbering of a cantilever plate.

The approximate solution utilizing the finite element method by applying a four-noded isoparametric element, in the following referred to PLANE4, is sought. The programming shall be done with MATLAB [2]

2 Isoparametric finite element PLANE4

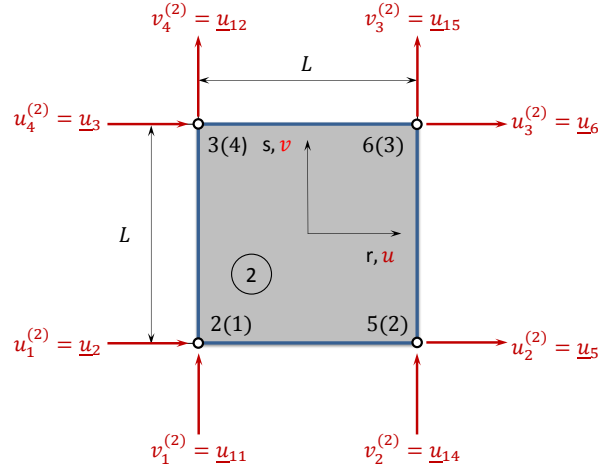


Figure 2: Four-noded quadrilateral element (number 2) in the global (x, y) and the natural (r, s) coordinate system.

2.1 Element coordinates and element displacements

Vector of element nodal coordinates

$$\mathbf{x}^{(e)} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix}, \quad \mathbf{y}^{(e)} = \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{Bmatrix}. \quad (1)$$

The global nodal coordinates are given by

$$\begin{aligned} \underline{\mathbf{x}}_{8 \times 1}^{(e)T} &= \{x_1 \quad x_2 \quad x_3 \quad x_4, \quad y_1 \quad y_2 \quad y_3 \quad y_4\} \\ &= \{\underline{x}_2 \quad \underline{x}_5 \quad \underline{x}_6 \quad \underline{x}_3 \quad \underline{x}_{11} \quad \underline{x}_{14} \quad \underline{x}_{15} \quad \underline{x}_{12}\}. \end{aligned} \quad (2)$$

Vector of element nodal displacements

$$\mathbf{u}^{(e)} = \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix}, \quad \mathbf{v}^{(e)} = \begin{Bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{Bmatrix}. \quad (3)$$

The global nodal displacements are given by

$$\begin{aligned} \underline{\mathbf{u}}_{8 \times 1}^{(e)T} &= \{u_1 \quad u_2 \quad u_3 \quad u_4, \quad v_1 \quad v_2 \quad v_3 \quad v_4\} \\ &= \{\underline{u}_2 \quad \underline{u}_5 \quad \underline{u}_6 \quad \underline{u}_3 \quad \underline{u}_{11} \quad \underline{u}_{14} \quad \underline{u}_{15} \quad \underline{u}_{12}\} \end{aligned} \quad (4)$$

Remark:

If the four nodes of the PLANE4 element for a particular discretization have the global node numbers $\{i, j, k, l\}$, then the global nodal displacements follow the numbering scheme according to

$$\underline{\mathbf{u}}_{8 \times 1}^{(e)T} = \left\{ \underline{u}_i \quad \underline{u}_j \quad \underline{u}_k \quad \underline{u}_l \quad \underline{u}_{i+n_{\text{Nodes}}} \quad \underline{u}_{j+n_{\text{Nodes}}} \quad \underline{u}_{k+n_{\text{Nodes}}} \quad \underline{u}_{l+n_{\text{Nodes}}} \right\}, \quad (5)$$

where n_{Nodes} denotes the number of nodes in the finite element model. For example, the finite element model given in Fig. 2 has $n_{\text{Nodes}} = 9$ nodes.

Relation between element coordinates and nodal coordinates

$$\begin{aligned} x^{(e)}(\mathbf{r}) &= \sum_{i=1}^4 h_i x_i^{(e)} = \mathbf{h} \mathbf{x}^{(e)}, \\ y^{(e)}(\mathbf{r}) &= \sum_{i=1}^4 h_i y_i^{(e)} = \mathbf{h} \mathbf{y}^{(e)}, \end{aligned} \quad (6)$$

with

$$\mathbf{r} = \begin{Bmatrix} r \\ s \end{Bmatrix}. \quad (7)$$

Relation between element displacements and nodal displacements (using the same interpolation functions as for element coordinates)

$$\begin{aligned} u^{(e)}(\mathbf{r}) &= \sum_{i=1}^4 h_i u_i^{(e)} = \mathbf{h} \mathbf{u}^{(e)}, \\ v^{(e)}(\mathbf{r}) &= \sum_{i=1}^4 h_i v_i^{(e)} = \mathbf{h} \mathbf{v}^{(e)}. \end{aligned} \quad (8)$$

The displacement interpolation functions are given by

$$\begin{aligned} h_1 &= \frac{1}{4}(1-r)(1-s), \\ h_2 &= \frac{1}{4}(1+r)(1-s), \\ h_3 &= \frac{1}{4}(1+r)(1+s), \\ h_4 &= \frac{1}{4}(1-r)(1+s), \end{aligned} \quad (9)$$

thus

$$\mathbf{h} = \{h_1, h_2, h_3, h_4\}. \quad (10)$$

Defining the element displacement vector according to

$$\mathbf{u}^{(e)}(\mathbf{r}) = \begin{Bmatrix} u^{(e)}(\mathbf{r}) \\ v^{(e)}(\mathbf{r}) \end{Bmatrix}, \quad (11)$$

(8) can be written according to

$$\mathbf{u}(\mathbf{r})_{2 \times 1}^{(e)} = \mathbf{H}(\mathbf{r})_{2 \times 8} \mathbf{u}_{8 \times 1}^{(e)}, \quad (12)$$

with

$$\mathbf{H}_{2 \times 8} = \begin{bmatrix} \mathbf{h}_{1 \times 4} & \mathbf{0}_{1 \times 4} \\ \mathbf{0}_{1 \times 4} & \mathbf{h}_{1 \times 4} \end{bmatrix}. \quad (13)$$

Derivatives of the displacement interpolation functions in natural coordinates are given by

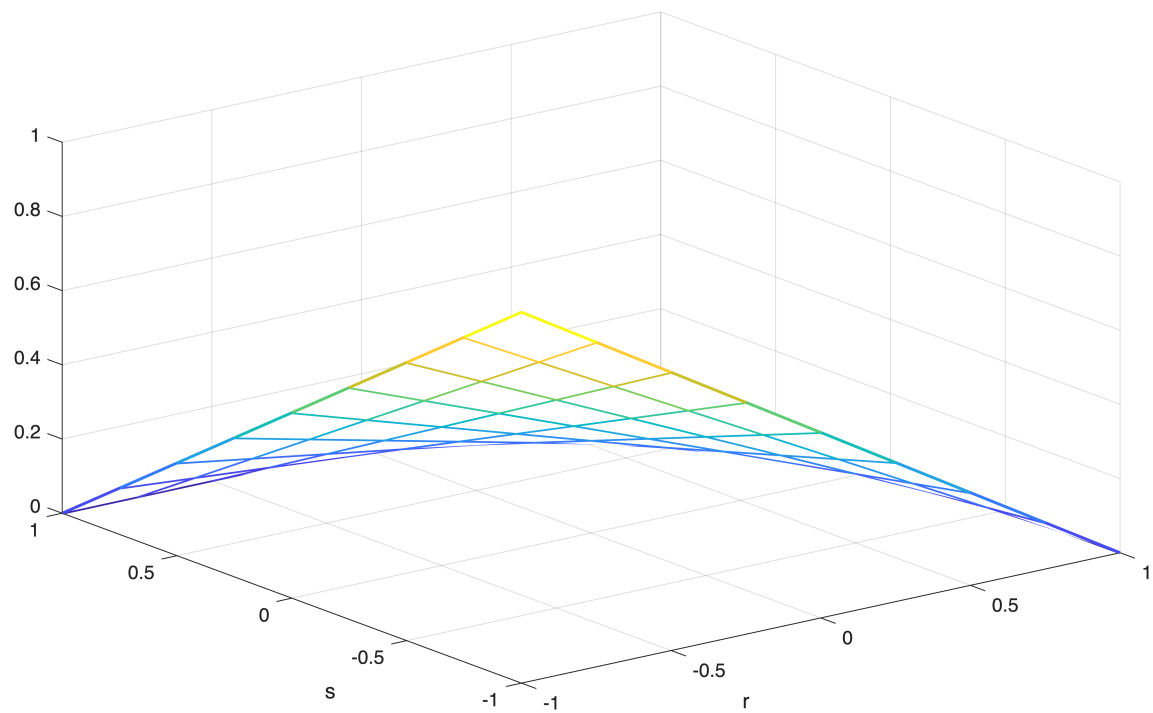
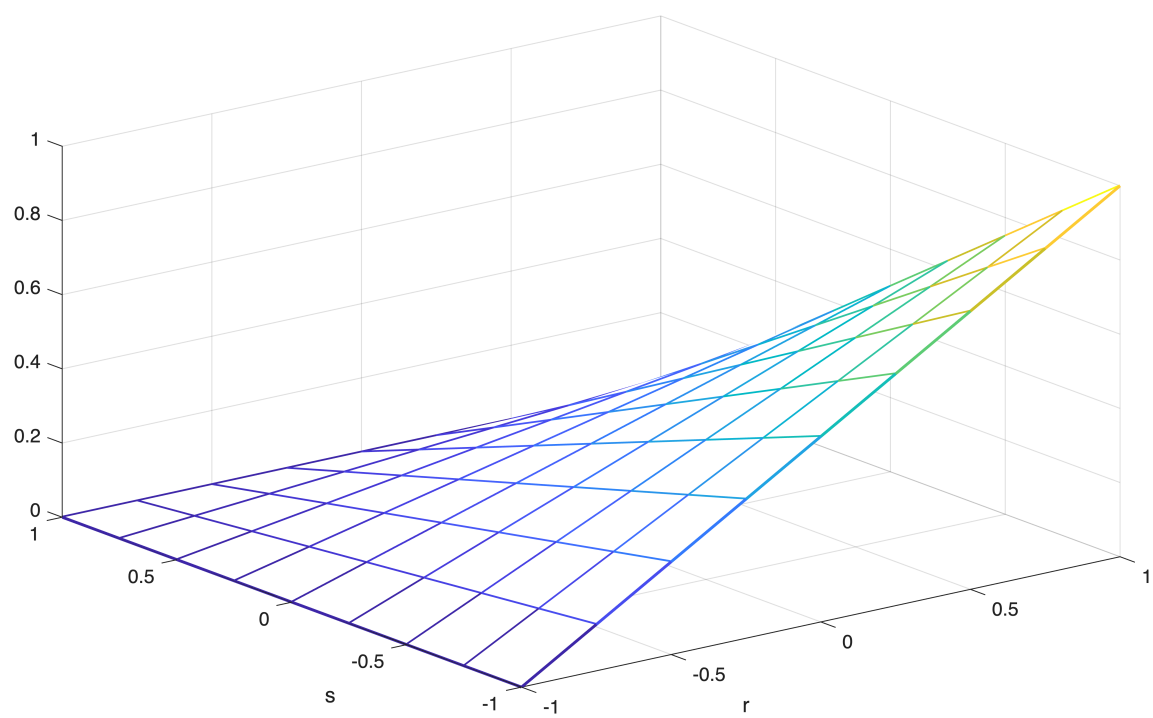
$$\frac{\partial \mathbf{h}}{\partial \mathbf{r}}_{2 \times 4} = \begin{bmatrix} \frac{\partial \mathbf{h}}{\partial r} \\ \frac{\partial \mathbf{h}}{\partial s} \end{bmatrix} = \begin{bmatrix} \frac{\partial h_1}{\partial r} & \frac{\partial h_2}{\partial r} & \frac{\partial h_3}{\partial r} & \frac{\partial h_4}{\partial r} \\ \frac{\partial h_1}{\partial s} & \frac{\partial h_2}{\partial s} & \frac{\partial h_3}{\partial s} & \frac{\partial h_4}{\partial s} \end{bmatrix}, \quad (14)$$

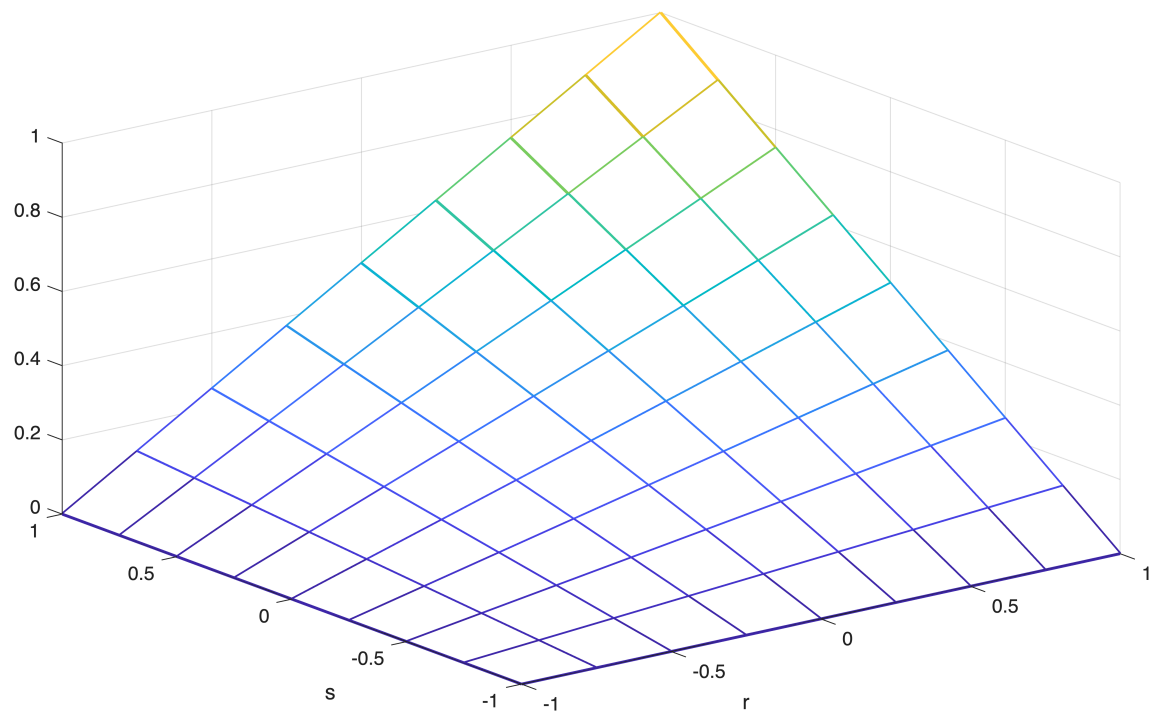
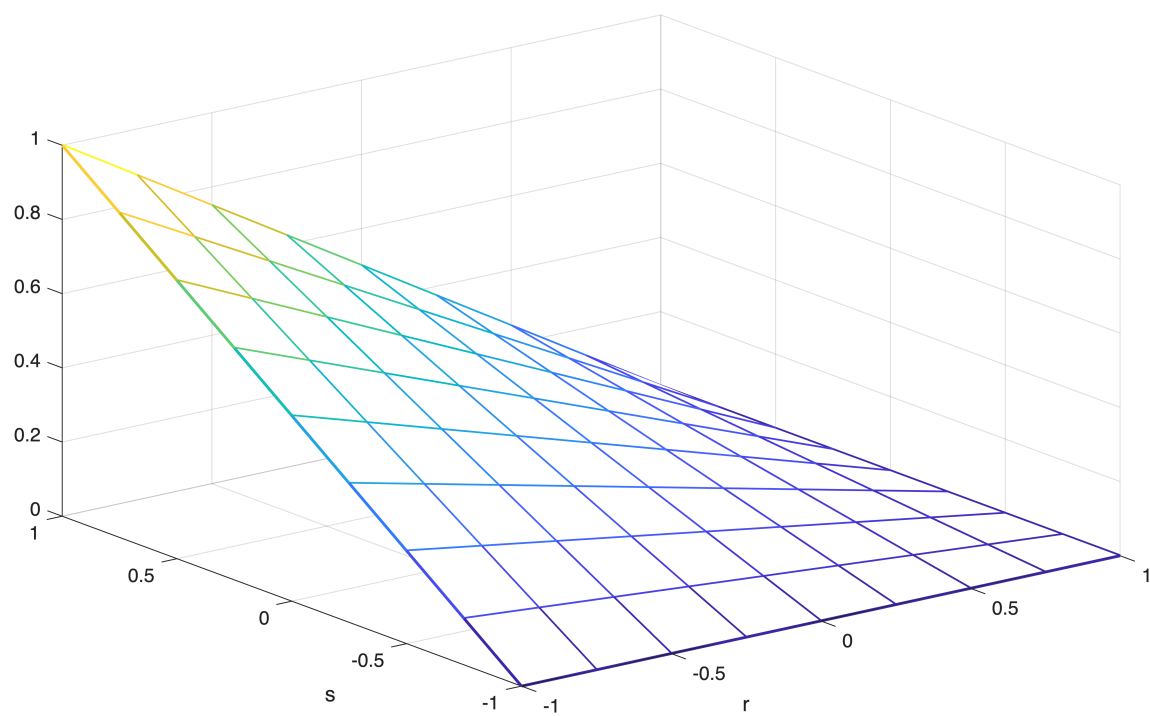
in particular

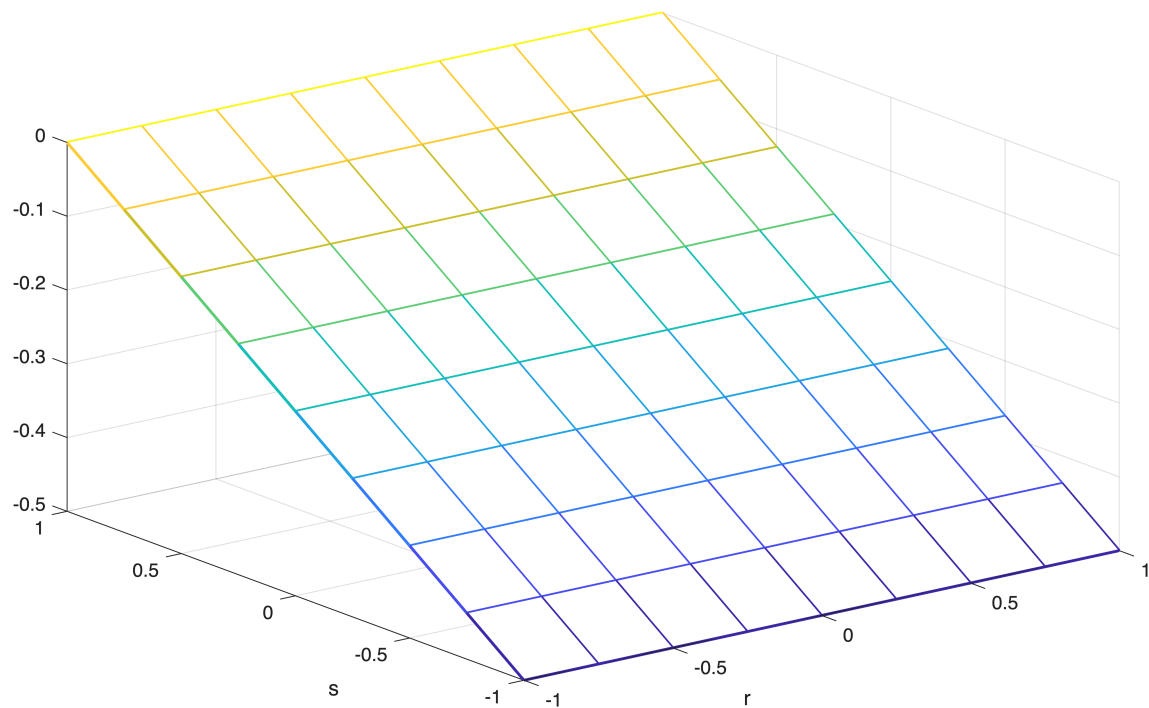
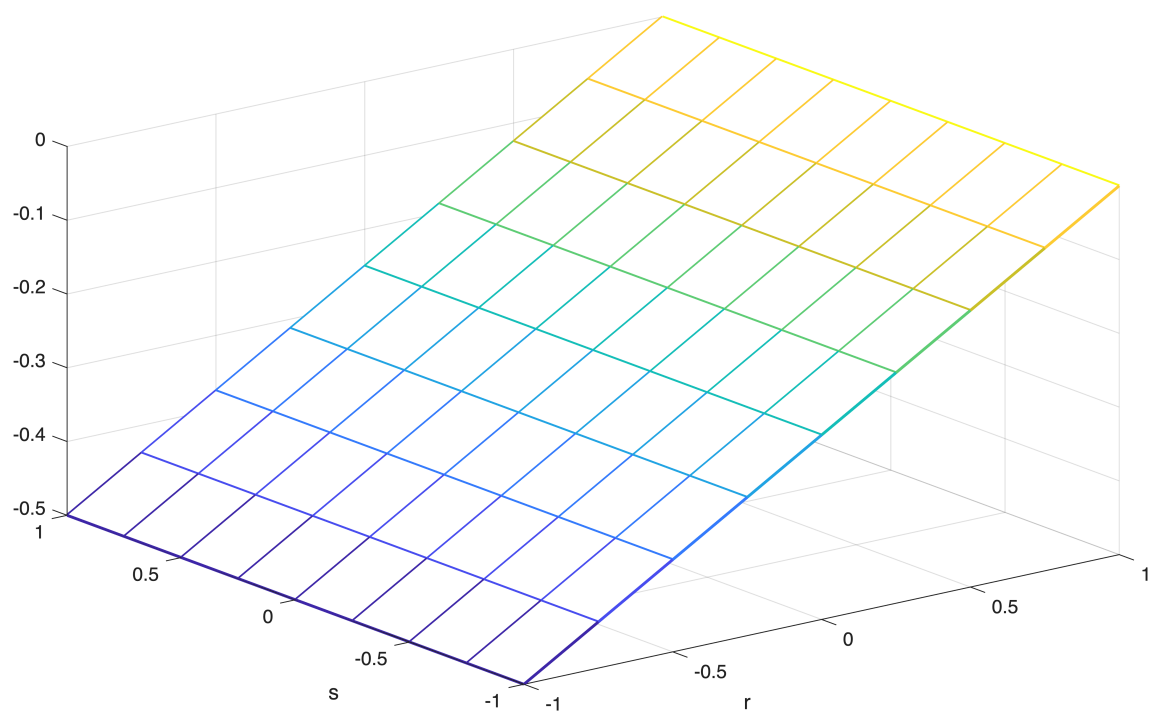
$$\begin{aligned} \frac{\partial h_1}{\partial r} &= -\frac{1}{4}(1-s), \\ \frac{\partial h_2}{\partial r} &= \frac{1}{4}(1-s), \\ \frac{\partial h_3}{\partial r} &= \frac{1}{4}(1+s), \\ \frac{\partial h_4}{\partial r} &= -\frac{1}{4}(1+s), \end{aligned} \quad (15)$$

and

$$\begin{aligned} \frac{\partial h_1}{\partial s} &= -\frac{1}{4}(1-r), \\ \frac{\partial h_2}{\partial s} &= -\frac{1}{4}(1+r), \\ \frac{\partial h_3}{\partial s} &= \frac{1}{4}(1+r), \\ \frac{\partial h_4}{\partial s} &= \frac{1}{4}(1-r). \end{aligned} \quad (16)$$

Figure 3: Shape function h_1 .Figure 4: Shape function h_2 .

Figure 5: Shape function h_3 .Figure 6: Shape function h_4 .

Figure 7: Derivative $\partial h_1 / \partial r$.Figure 8: Derivative $\partial h_1 / \partial s$.

2.2 Element strains

For plane stress, the displacement field can be expressed by

$$\mathbf{u}(x, y) = \begin{Bmatrix} u(x, y) \\ v(x, y) \end{Bmatrix}, \quad (17)$$

i.e. the deformation of each structural point is given by its horizontal and vertical components u and v , respectively.

The corresponding strains are given by

$$\boldsymbol{\epsilon}^T = \{\epsilon_{xx} \quad \epsilon_{yy} \quad \gamma_{xy}\} \quad (18)$$

with

$$\epsilon_{xx} = \frac{\partial u}{\partial x}, \quad \epsilon_{yy} = \frac{\partial v}{\partial y} \quad (19)$$

and

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}. \quad (20)$$

In order to calculate the derivatives of the displacement interpolation functions with respect to the global coordinate system, i.e.

$$\frac{\partial \mathbf{h}}{\partial \mathbf{x}_{2 \times 4}} = \frac{\partial \mathbf{r}}{\partial \mathbf{x}_{2 \times 2}} \frac{\partial \mathbf{h}}{\partial \mathbf{r}_{2 \times 4}}, \quad (21)$$

the inverse of the Jacobi matrix is needed because of

$$\mathbf{J}^{-1} = \frac{\partial \mathbf{r}}{\partial \mathbf{x}}. \quad (22)$$

The Jacobi matrix is given by

$$\mathbf{J} = \frac{\partial \mathbf{x}}{\partial \mathbf{r}} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial s} & \frac{\partial y}{\partial s} \end{bmatrix} \quad (23)$$

where the elements of $\mathbf{J}$ are given by

$$\begin{aligned}\frac{\partial x}{\partial r} &= \sum_{i=1}^4 \frac{\partial h_i}{\partial r} x_i, \\ \frac{\partial x}{\partial s} &= \sum_{i=1}^4 \frac{\partial h_i}{\partial s} x_i, \\ \frac{\partial y}{\partial r} &= \sum_{i=1}^4 \frac{\partial h_i}{\partial r} y_i, \\ \frac{\partial y}{\partial s} &= \sum_{i=1}^4 \frac{\partial h_i}{\partial s} y_i.\end{aligned}\tag{24}$$

Thus the Jacobi matrix can be determined by

$$\mathbf{J} = \frac{\partial \mathbf{h}}{\partial \mathbf{r}} [\mathbf{x}^{(e)}, \mathbf{y}^{(e)}]_{4 \times 2}.\tag{25}$$

Finally, by introducing the strain-displacement matrix $\mathbf{B}$, the element strains $\boldsymbol{\epsilon}$ can be written according to

$$\boldsymbol{\epsilon}_{3 \times 1}^{(e)} = \mathbf{B}_{3 \times 8} \mathbf{u}_{8 \times 1}^{(e)},\tag{26}$$

with

$$\mathbf{B} = \begin{bmatrix} \frac{\partial \mathbf{h}}{\partial x} & \mathbf{0} \\ \mathbf{0} & \frac{\partial \mathbf{h}}{\partial y} \\ \frac{\partial \mathbf{h}}{\partial y} & \frac{\partial \mathbf{h}}{\partial x} \end{bmatrix}.\tag{27}$$

The stress vector

$$\boldsymbol{\tau}^T = \{\tau_{xx} \quad \tau_{yy} \quad \tau_{xy}\}\tag{28}$$

for linear elastic material is defined by

$$\boldsymbol{\tau} = \mathbf{C} \boldsymbol{\epsilon},\tag{29}$$

where the corresponding stress-strain relation is given by

$$\begin{Bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{xy} \end{Bmatrix} = \frac{E}{1 - \mu^2} \begin{bmatrix} 1 & \mu & 0 \\ \mu & 1 & 0 \\ 0 & 0 & \frac{1 - \mu}{2} \end{bmatrix} \begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{xy} \end{Bmatrix}.\tag{30}$$

The equilibrium equations for plane stress are given by

$$\begin{aligned}\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + f_x &= 0, \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + f_y &= 0,\end{aligned}\tag{31}$$

which corresponds to the class of elliptic partial differential equations.

2.3 Stiffness matrix

The element stiffness for element e is given by

$$\mathbf{K}^{(e)} = \int_{V^{(e)}} \mathbf{B}^T \mathbf{C} \mathbf{B} dV^{(e)}.\tag{32}$$

For the quadrilateral element with uniform thickness t , i.e. $dV = t \det \mathbf{J} dr ds$, it follows that

$$\mathbf{K}^{(e)} = t \int_{A^{(e)}} \mathbf{B}^T(\mathbf{r}) \mathbf{C} \mathbf{B}(\mathbf{r}) \det \mathbf{J} d\mathbf{r}.\tag{33}$$

3 Numerical integration

The element stiffness matrix given by (32) is numerically integrated by applying Gauss quadrature. Therefore, (33) can be written as

$$\mathbf{K}^{(e)} = t \int_{A^{(e)}} \mathbf{D} d\mathbf{r},\tag{34}$$

where

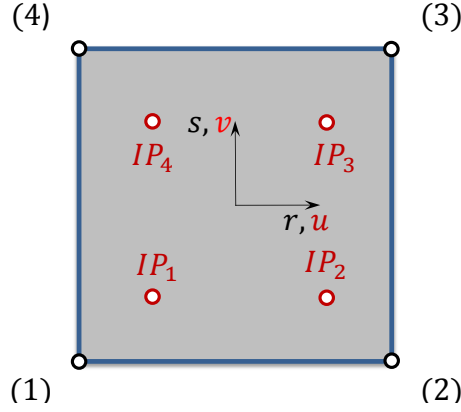
$$\mathbf{D} = \mathbf{B}^T(\mathbf{r}) \mathbf{C} \mathbf{B}(\mathbf{r}) \det \mathbf{J}.\tag{35}$$

Using Gauss quadrature, (34) can be approximated according to

$$\mathbf{K}^{(e)} \approx t \sum_i \sum_j \alpha_{ij} \mathbf{D}_{ij}.\tag{36}$$

Hereby $\mathbf{D}_{ij}$ is the matrix $\mathbf{D}$ evaluated at point $r_{ij} = (r_i, s_j)$ and α_{ij} is a given weighting factor that depends on the value of r_i .

According to Fig. 9, the quadrilateral four-noded isoparametric element is usually integrated using four integration points.

Figure 9: 2×2 integration for four-noded quadrilateral element .

These integration points have the following locations in the natural coordinate system:

$$\begin{aligned} \text{IP}_1: r_{11} = (r_1, s_1) &= (-1/\sqrt{3}, -1/\sqrt{3}), \alpha_{11} = 1 \\ \text{IP}_2: r_{12} = (r_1, s_2) &= (1/\sqrt{3}, -1/\sqrt{3}), \alpha_{12} = 1 \\ \text{IP}_3: r_{21} = (r_2, s_1) &= (1/\sqrt{3}, 1/\sqrt{3}), \alpha_{21} = 1 \\ \text{IP}_4: r_{22} = (r_2, s_2) &= (-1/\sqrt{3}, 1/\sqrt{3}), \alpha_{22} = 1 \end{aligned}$$

4 Imposition of boundary conditions

Next the imposition of boundary conditions is shown, see also [1]. For a static system, the equilibrium equations can be written as

$$\begin{bmatrix} \mathbf{K}_{aa} & \mathbf{K}_{ab} \\ \mathbf{K}_{ba} & \mathbf{K}_{bb} \end{bmatrix} \begin{Bmatrix} \mathbf{u}_a \\ \mathbf{u}_b \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}_a \\ \mathbf{f}_b \end{Bmatrix} \quad (37)$$

where $\mathbf{u}_b$ are the known (prescribed) displacements. By solving for $\mathbf{u}_a$ one obtains

$$\mathbf{K}_{aa}\mathbf{u}_a = \mathbf{f}_a - \mathbf{K}_{ab}\mathbf{u}_b. \quad (38)$$

Once the displacements $\mathbf{u}_a$ have been calculated from (38), the reactions can be calculated according to

$$\mathbf{f}_b = \mathbf{K}_{ba}\mathbf{u}_a + \mathbf{K}_{bb}\mathbf{u}_b. \quad (39)$$

5 Nodal point forces

The element nodal point forces can be calculated according to

$$\mathbf{r}^{(e)} = - \int_{V^{(e)}} \mathbf{B}^{(e)T} \boldsymbol{\tau}^{(e)} dV^{(e)}, \quad (40)$$

where $\boldsymbol{\tau}^{(e)} = \mathbf{C}^{(e)} \boldsymbol{\epsilon}^{(e)}$.

6 Results

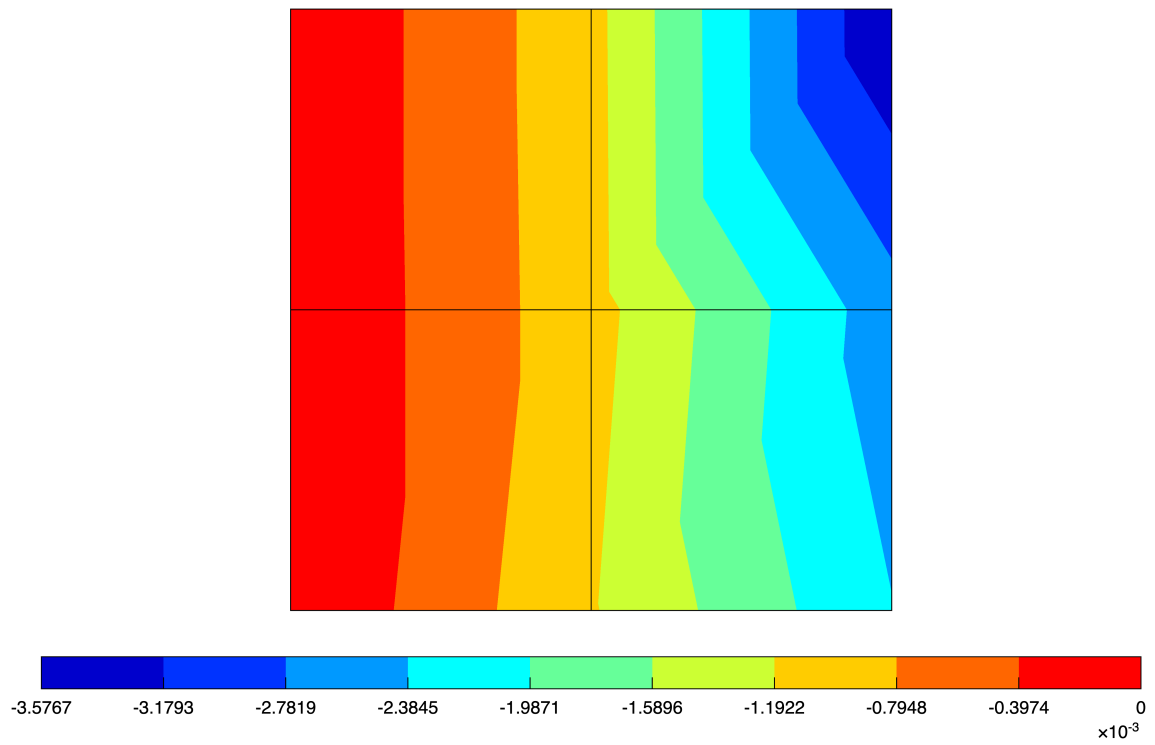
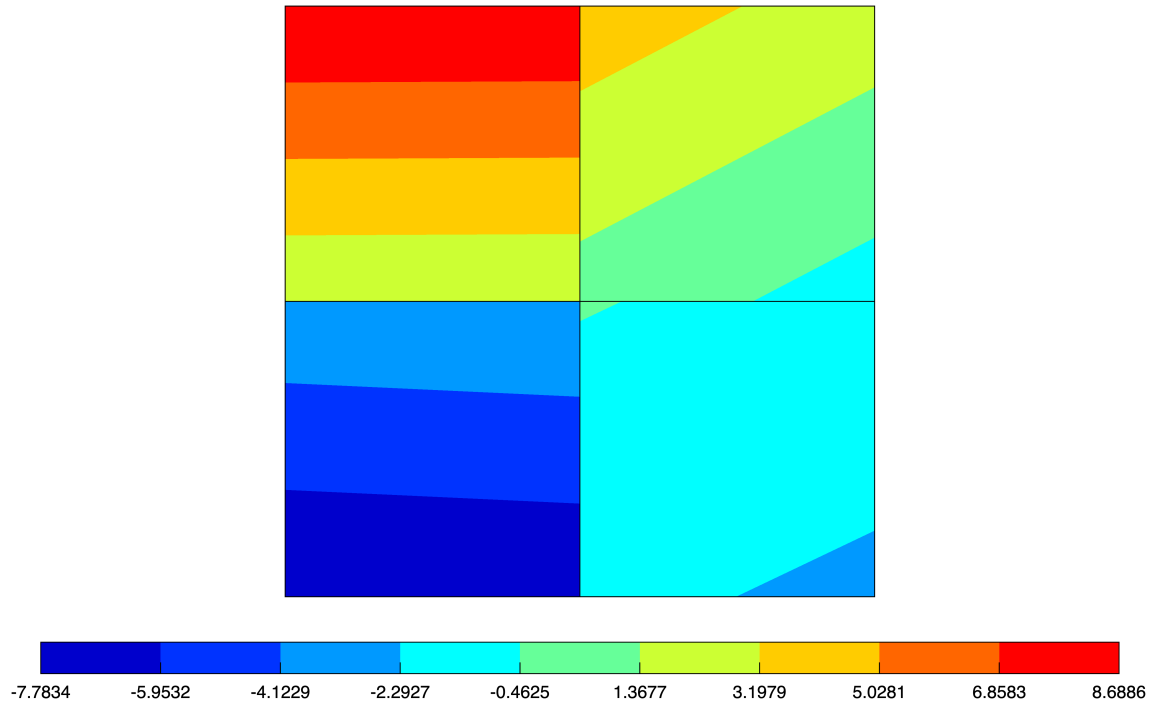
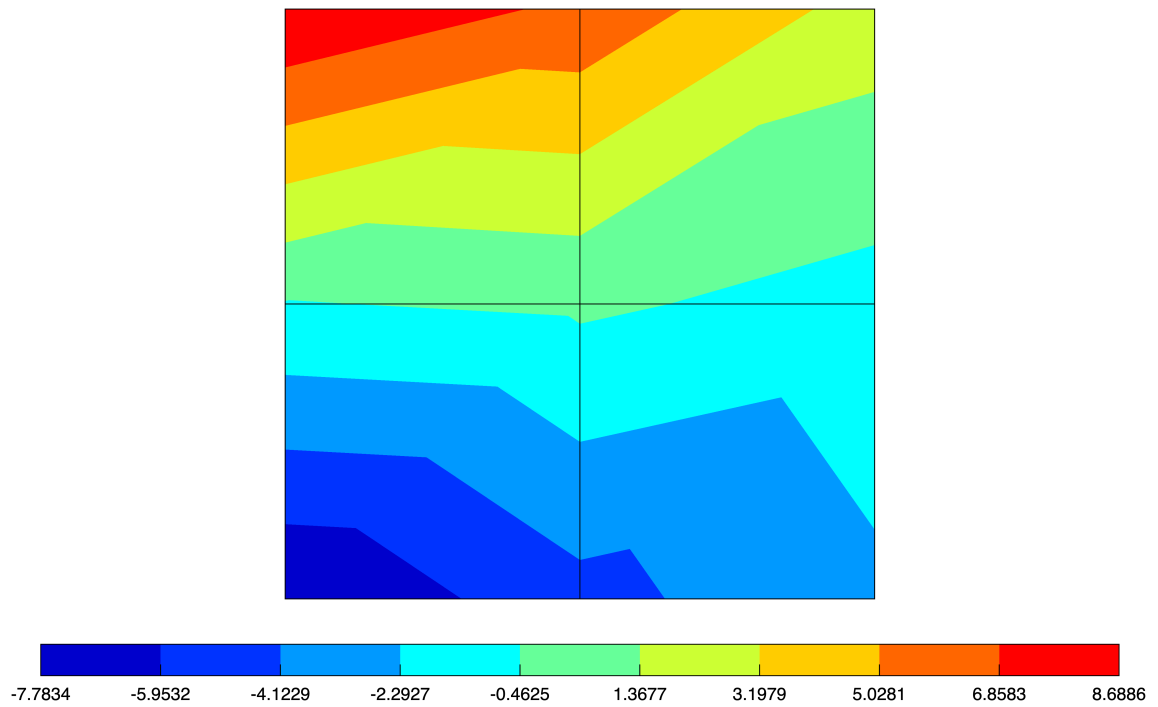
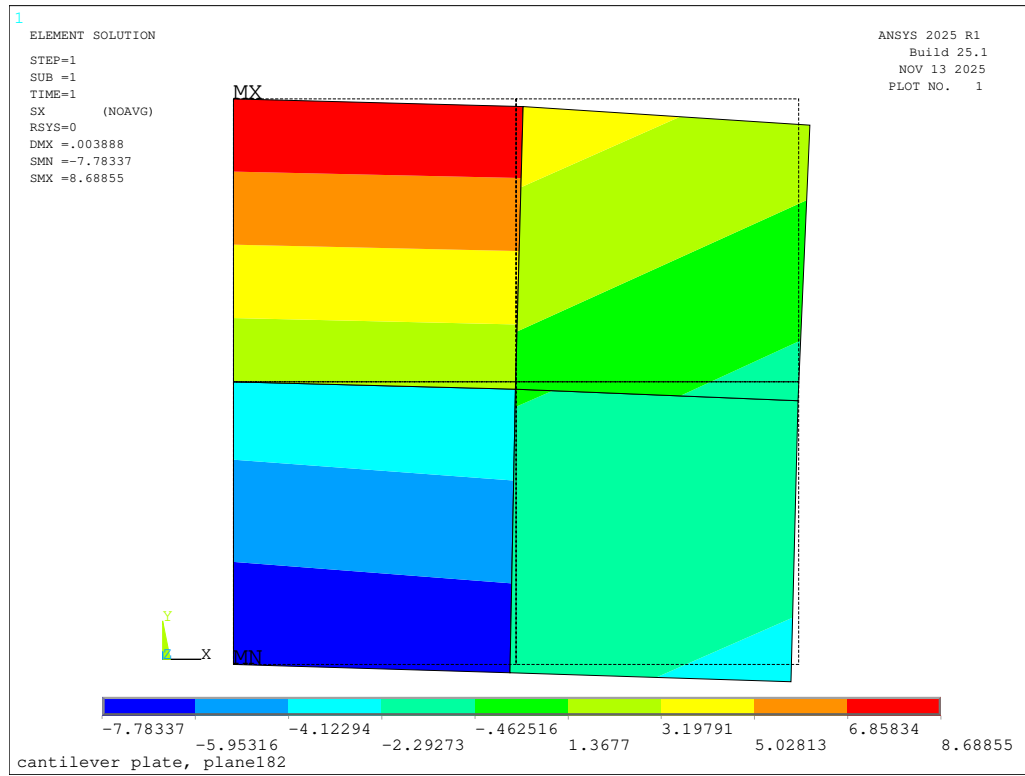


Figure 10: MATLAB solution for vertical displacements.

Figure 11: MATLAB solution for element normal stresses in x -direction.Figure 12: MATLAB solution for nodal normal stresses in x -direction.

Figure 13: ANSYS solution for element normal stresses in x -direction.

Listing 1: MATLAB results

```

1 Displacements: 16-Oct-2025 21:55:26
2
3   Node      UX      UY
4     1  0.0000e+00  0.0000e+00
5     2  0.0000e+00  0.0000e+00
6     3  0.0000e+00  0.0000e+00
7     4 -8.4250e-04 -1.1584e-03
8     5 -8.3127e-05 -1.0407e-03
9     6  9.7837e-04 -1.0572e-03
10    7 -1.0684e-03 -2.3698e-03
11    8 -7.7190e-05 -2.6200e-03
12    9  1.5241e-03 -3.5767e-03
13
14 Maximum absolute values
15 Node      9      9
16 Value     1.5241e-03 -3.5767e-03

```

Listing 2: ANSYS results

```

1 ***** POST1 NODAL DEGREE OF FREEDOM LISTING *****
2
3 LOAD STEP=      1  SUBSTEP=      1
4 TIME=      1.0000  LOAD CASE=      0
5
6 THE FOLLOWING DEGREE OF FREEDOM RESULTS ARE IN THE GLOBAL COORDINATE SYSTEM
7
8   NODE      UX      UY      UZ      USUM
9     1  0.0000  0.0000  0.0000  0.0000
10    2  0.0000  0.0000  0.0000  0.0000
11    3  0.0000  0.0000  0.0000  0.0000
12    4 -0.84250E-03-0.11584E-02  0.0000  0.14324E-02
13    5 -0.83127E-04-0.10407E-02  0.0000  0.10440E-02
14    6  0.97837E-03-0.10572E-02  0.0000  0.14404E-02
15    7 -0.10684E-02-0.23698E-02  0.0000  0.25995E-02
16    8 -0.77190E-04-0.26200E-02  0.0000  0.26212E-02
17    9  0.15241E-02-0.35767E-02  0.0000  0.38879E-02
18
19 MAXIMUM ABSOLUTE VALUES
20 NODE      9      9      0      9
21 VALUE     0.15241E-02-0.35767E-02  0.0000  0.38879E-02

```


Listing 3: MATLAB results

```

1 Element Stresses: 16-Oct-2025 21:55:26
2
3
4 Element 1:
5 Node      X      Y      XY
6   1 -7.7834e+00 -2.0738e+00 -3.9298e+00
7   4 -7.5481e+00 -1.2895e+00 -2.1592e+00
8   5 -2.4894e+00  0.22812e+00 -1.8847e+00
9   2 -2.7247e+00 -0.55616e+00 -3.6553e+00
10
11 Element 2:
12 Node      X      Y      XY
13   2  1.6171e+00  0.44857e+00 -3.3110e+00
14   5  1.5842e+00  0.33879e+00 -0.83603e+00
15   6  8.6556e+00  2.4602e+00 -0.87445e+00
16   3  8.6886e+00  2.5700e+00 -3.3495e+00
17
18 Element 3:
19 Node      X      Y      XY
20   4 -1.9029e+00 -0.15115e+00 -1.9416e+00
21   7 -2.6383e+00 -2.6023e+00 -1.4010e+00
22   8 -1.0938e+00 -2.1389e+00 -2.2589e+00
23   5 -0.35845e+00 0.31220e+00 -2.7995e+00
24
25 Element 4:
26 Node      X      Y      XY
27   5  0.63990e+00 -2.0672e+00 -2.4329e+00
28   8 -1.2391e+00 -8.3304e+00 -1.1744e+00
29   9  2.3568e+00 -7.2517e+00 -3.3665e+00
30   6  4.2358e+00 -0.98846e+00 -4.6251e+00

```

Listing 4: ANSYS results

```

1  NODE      SX      SY      SZ      SXY      SYZ      SXZ
2    1 -7.7834 -2.0738  0.0000 -3.9298  0.0000  0.0000
3    4 -7.5481 -1.2895  0.0000 -2.1592  0.0000  0.0000
4    5 -2.4894  0.22812  0.0000 -1.8847  0.0000  0.0000
5    2 -2.7247 -0.55616  0.0000 -3.6553  0.0000  0.0000
6
7  ELEMENT=      2      PLANE182
8  NODE      SX      SY      SZ      SXY      SYZ      SXZ
9    2  1.6171  0.44857  0.0000 -3.3110  0.0000  0.0000
10   5  1.5842  0.33879  0.0000 -0.83603  0.0000  0.0000
11   6  8.6556  2.4602  0.0000 -0.87445  0.0000  0.0000
12   3  8.6886  2.5700  0.0000 -3.3495  0.0000  0.0000
13
14  ELEMENT=      3      PLANE182
15  NODE      SX      SY      SZ      SXY      SYZ      SXZ
16   4 -1.9029 -0.15115  0.0000 -1.9416  0.0000  0.0000
17   7 -2.6383 -2.6023  0.0000 -1.4010  0.0000  0.0000
18   8 -1.0938 -2.1389  0.0000 -2.2589  0.0000  0.0000
19   5 -0.35845 0.31220  0.0000 -2.7995  0.0000  0.0000
20
21  ELEMENT=      4      PLANE182
22  NODE      SX      SY      SZ      SXY      SYZ      SXZ
23   5  0.63990 -2.0672  0.0000 -2.4329  0.0000  0.0000
24   8 -1.2391 -8.3304  0.0000 -1.1744  0.0000  0.0000
25   9  2.3568 -7.2517  0.0000 -3.3665  0.0000  0.0000
26   6  4.2358 -0.98846  0.0000 -4.6251  0.0000  0.0000

```

7 File listing

7.1 ANSYS

Listing 5: ANSYS code cantilever_plate.inp

```

1 finish
2 /clear
3
4 /filename,,1
5 !-----
6 /prep7
7 /title,cantilever plate, plane182
8
9 /triad,lbot
10
11 et,1,plane182,0 ! Bbar
12 mp,ex,1,2.1E5
13 mp,prxy,1,0.3
14
15 Le = 40
16 He = 40
17 Wi = 1
18 Fo = 100
19
20 I=0
21 *do,X,0,Le,Le/2
22     n,I+1,X,0,0
23     n,I+2,X,He/2,0
24     n,I+3,X,He,0
25     I=I+3
26 *enddo
27
28 e,1,4,5,2
29 e,2,5,6,3
30 e,4,7,8,5
31 e,5,8,9,6
32
33 nsel,s,loc,x,0
34 d,all,all
35 nsel,all
36
37 f,9,fy,-Fo/Wi
38
39 /pnum,node,1
40 /pnum,elem,1
41 eplot
42
43 finish
44
45 !-----
46 /solu
47 eresx,no
48 solve
49 finish
50
51 !-----
52 /post1
53
54 ! element results
55 presol,s
56 plesol,s,x,1
57
58 ! plvect,S,1,,,vect,node
59 finish

```

List of Figures

1	Mesh, node and element numbering of a cantilever plate.	1
2	Four-noded quadrilateral element (number 2) in the global (x, y) and the natural (r, s) coordinate system.	2
3	Shape function h_1	5
4	Shape function h_2	5
5	Shape function h_3	6
6	Shape function h_4	6
7	Derivative $\partial h_1 / \partial r$	7
8	Derivative $\partial h_1 / \partial s$	7
9	2×2 integration for four-noded quadrilateral element	11
10	MATLAB solution for vertical displacements.	12
11	MATLAB solution for element normal stresses in x -direction.	13
12	MATLAB solution for nodal normal stresses in x -direction.	13
13	ANSYS solution for element normal stresses in x -direction.	14

References

- [1] K.-J. Bathe. *Finite element procedures*. Prentice Hall, Upper Saddle River, New Jersey, 1996.
- [2] MATLAB. *Release 2025a*. The MathWorks, Inc. 2025.